

Aplicar filtros a consultas SQL

Aplicar filtros a consultas SQL

Descripción del proyecto

Este proyecto demuestra el uso de consultas SQL (Structured Query Language) para analizar registros del sistema e investigar posibles incidentes de seguridad. El objetivo fundamental es aplicar filtros lógicos para auditar los accesos de los usuarios y gestionar la información organizativa, asegurando que se puedan detectar anomalías de manera eficiente. Al dominar estas consultas, mejoramos la capacidad de monitoreo, facilitamos la respuesta ante incidentes y fortalecemos el control sobre la información crítica de la base de datos.

Recuperar intentos de inicio de sesión fallidos fuera del horario laboral

Para comenzar la investigación de posibles actividades anómalas, extraje los registros de intentos de acceso fallidos ocurridos después del horario comercial (18:00). Primero, utilicé `SELECT *` para seleccionar todos los datos de la tabla `log_in_attempts`. Luego, apliqué una cláusula `WHERE` con un operador `AND` para filtrar los resultados. La primera condición, `login_time > '18:00'`, filtra los intentos posteriores a las 18:00, y la segunda condición, `success = FALSE`, aísla los inicios de sesión fallidos. Esto garantiza precisión al investigar eventos de alto riesgo.

```
MariaDB [organization]> SELECT *  
-> FROM log_in_attempts  
-> WHERE login_time > '18:00' AND success = FALSE;
```

event_id	username	login_date	login_time	country	ip_address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
52	cjackson	2022-05-10	22:07:07	CAN	192.168.58.57	0
69	wjaffrey	2022-05-11	19:55:15	USA	192.168.100.17	0
82	abernard	2022-05-12	23:38:46	MEX	192.168.234.49	0
87	apatel	2022-05-08	22:38:31	CANADA	192.168.132.153	0
96	ivelasco	2022-05-09	22:36:36	CAN	192.168.84.194	0
104	asundara	2022-05-11	18:38:07	US	192.168.96.200	0
107	bisles	2022-05-12	20:25:57	USA	192.168.116.187	0
111	aestrada	2022-05-10	22:00:26	MEXICO	192.168.76.27	0
127	abellmas	2022-05-09	21:20:51	CANADA	192.168.70.122	0
131	bisles	2022-05-09	20:03:55	US	192.168.113.171	0
155	cgriffin	2022-05-12	22:18:42	USA	192.168.236.176	0
160	jclark	2022-05-10	20:49:00	CANADA	192.168.214.49	0
199	yappiah	2022-05-11	19:34:48	MEXICO	192.168.44.232	0

19 rows in set (0.015 sec)

Figura 1. Visualización de intentos de inicio de sesión fallidos ocurridos después de las 18:00 horas utilizando la cláusula `AND`.

Recuperar intentos de inicio de sesión en fechas específicas

Con el propósito de auditar la actividad del sistema durante un posible incidente, revisé los inicios de sesión de dos días específicos. Empecé seleccionando todos los datos de la tabla `log_in_attempts`. Luego, utilicé la cláusula `WHERE` con el operador `OR` para filtrar la consulta: `WHERE login_date = '2022-05-09' OR login_date = '2022-05-08'`; . La primera condición filtra los accesos del 2022-05-09 y la segunda los del 2022-05-08. El operador `OR` es indispensable aquí porque devuelve los registros que coinciden con al menos una de las fechas especificadas.

```
MariaDB [organization]> SELECT *  
-> FROM log_in_attempts  
-> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
24	arusso	2022-05-09	06:49:39	MEXICO	192.168.171.192	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1
26	apatel	2022-05-08	17:27:00	CANADA	192.168.123.105	1
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
30	yappiah	2022-05-09	03:22:22	MEX	192.168.124.48	1
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
36	asundara	2022-05-08	09:00:42	US	192.168.78.151	1
38	sbaelish	2022-05-09	14:40:01	USA	192.168.60.42	1
39	yappiah	2022-05-09	07:56:40	MEXICO	192.168.57.115	1
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
43	mcouliba	2022-05-08	02:35:34	CANADA	192.168.16.208	0
44	daquino	2022-05-08	07:02:35	CANADA	192.168.168.144	0
47	dkot	2022-05-08	05:06:45	US	192.168.233.24	1
49	asundara	2022-05-08	14:00:01	US	192.168.173.213	0
53	nmason	2022-05-08	11:51:38	CAN	192.168.133.188	1
56	acook	2022-05-08	04:56:30	CAN	192.168.209.130	1
58	ivelasco	2022-05-09	17:20:54	CAN	192.168.57.162	0
61	dtanaka	2022-05-09	09:45:18	USA	192.168.98.221	1
65	aalonso	2022-05-09	23:42:12	MEX	192.168.52.37	1
66	aestrada	2022-05-08	21:58:32	MEX	192.168.67.223	1
67	abernard	2022-05-09	11:53:41	MEX	192.168.118.29	1
68	mrh	2022-05-08	17:16:13	US	192.168.42.248	1
70	tmitchel	2022-05-09	10:55:17	MEXICO	192.168.87.199	1
71	mcouliba	2022-05-09	06:57:42	CAN	192.168.55.169	0
72	alevitsk	2022-05-08	12:09:10	CANADA	192.168.139.176	1

Figura 2. Extracción de registros de acceso para fechas consecutivas empleando el operador lógico `OR`.

Recuperar intentos de inicio de sesión fuera de México

Como parte del monitoreo de accesos internacionales, realicé un filtrado para identificar tráfico originado fuera de México. Inicié seleccionando todos los datos de `log_in_attempts`.

Después, utilicé una cláusula `WHERE` con el operador `NOT` para filtrar los países que no sean México: `WHERE NOT country LIKE 'MEX%'`; . Utilicé `LIKE` con el patrón `MEX%` porque la base de datos representa a México como `MEX` y `MEXICO`. El signo de porcentaje (%) funciona como comodín para abarcar cualquier número de caracteres, excluyendo así eficazmente todo el tráfico de esa región.

```
MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE NOT country LIKE 'MEX%';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
7	eraab	2022-05-11	01:45:14	CAN	192.168.170.243	1
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
10	jrafael	2022-05-12	09:33:19	CANADA	192.168.228.221	0
11	sgilmore	2022-05-11	10:16:29	CANADA	192.168.140.81	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
13	mrh	2022-05-11	09:29:34	USA	192.168.246.135	1
14	sbaelish	2022-05-10	10:20:18	US	192.168.16.99	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
17	pwashing	2022-05-11	02:33:02	USA	192.168.81.89	1
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
19	jhill	2022-05-12	13:09:04	US	192.168.142.245	1
21	iuduike	2022-05-11	17:50:00	US	192.168.131.147	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1
26	apatel	2022-05-08	17:27:00	CANADA	192.168.123.105	1
29	bisles	2022-05-11	01:21:22	US	192.168.85.186	0
31	acook	2022-05-12	17:36:45	CANADA	192.168.58.232	0
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
33	zbernal	2022-05-11	02:52:10	US	192.168.72.59	1
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
36	asundara	2022-05-08	09:00:42	US	192.168.78.151	1
37	eraab	2022-05-10	06:03:41	CANADA	192.168.152.148	0
38	sbaelish	2022-05-09	14:40:01	USA	192.168.60.42	1
41	apatel	2022-05-10	17:39:42	CANADA	192.168.46.207	0
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0

Figura 3. Filtrado de eventos de inicio de sesión excluyendo el tráfico originado en México mediante los operadores `NOT` y `LIKE`.

Recuperar empleados en Marketing

Para actualizar los equipos del personal de Marketing en la región Este, necesité aislar a estos empleados. Seleccioné los datos de la tabla employees y utilicé la cláusula WHERE con el operador AND: WHERE department = 'Marketing' AND office LIKE 'East%';. La primera condición, department = 'Marketing', filtra al personal de ese departamento. La segunda condición, office LIKE 'East%', utiliza el comodín % para coincidir con cualquier oficina que comience con la palabra "East" (por ejemplo, East-170 o East-195), proporcionando un listado exacto.

```
MariaDB [organization]> SELECT *  
-> FROM employees  
-> WHERE department = 'Marketing' AND office LIKE 'East%';  
+-----+-----+-----+-----+-----+  
| employee_id | device_id | username | department | office |  
+-----+-----+-----+-----+-----+  
|          1000 | a320b137c219 | elarson | Marketing | East-170 |  
|          1052 | a192b174c940 | jdarosa | Marketing | East-195 |  
|          1075 | x573y883z772 | fbautist | Marketing | East-267 |  
|          1088 | k865l965m233 | rgosh | Marketing | East-157 |  
|          1103 | NULL | randerss | Marketing | East-460 |  
|          1156 | a184b775c707 | dellery | Marketing | East-417 |  
|          1163 | h679i515j339 | cwilliam | Marketing | East-216 |  
+-----+-----+-----+-----+-----+  
7 rows in set (0.000 sec)
```

Figura 4. Visualización del personal del departamento de Marketing situado en las oficinas de la región Este.

Recuperar empleados en Finanzas o Ventas

Para desplegar una actualización de seguridad específica, requerí la lista de usuarios de las áreas de Finanzas y Ventas. Comencé consultando la tabla employees. Luego, apliqué la cláusula WHERE con el operador OR: WHERE department = 'Finance' OR department = 'Sales' ;. Utilicé OR en lugar de AND porque necesito a los empleados que pertenezcan a cualquiera de los dos departamentos. La primera condición filtra a los de Finanzas (department = 'Finance') y la segunda a los de Ventas (department = 'Sales'), unificando eficazmente el listado.

```
MariaDB [organization]> SELECT *  
-> FROM employees  
-> WHERE department = 'Finance' OR department = 'Sales';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1015	p611q262r945	jsoto	Finance	North-271
1017	r550s824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	iuduike	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1029	d336e475f676	ivelasco	Finance	East-156
1035	j236k303l245	bisles	Sales	South-171
1039	n253o917p623	cjackson	Sales	East-378
1041	p929q222r778	cgriffin	Sales	North-208
1044	s429t157u159	tbarnes	Finance	West-415
1045	t567u844v434	pwashing	Finance	East-115
1046	u429v921w138	daquino	Finance	West-280
1047	v109w587x644	cward	Finance	West-373
1048	w167x592y375	tmitchel	Finance	South-288
1049	NULL	jreckley	Finance	Central-295
1050	y132z930a114	csimmons	Finance	North-468
1057	f370g535h632	mscott	Sales	South-270
1062	k367l639m697	redwards	Finance	North-180
1063	l686m140n569	lpope	Sales	East-226
1066	o678p794q957	ttyrell	Sales	Central-444
1069	NULL	jpark	Finance	East-110
1071	t244u829v723	zdutchma	Sales	West-348
1072	u905v920w694	esmith	Sales	East-421

Figura 5. Extracción combinada de los perfiles de empleados correspondientes a los departamentos de Finanzas y Ventas.

Recuperar todos los empleados que no están en TI

Para una actualización de seguridad dirigida a usuarios ajenos a TI, necesité excluir al personal de ese departamento. Tras seleccionar los datos de la tabla employees, utilicé la cláusula WHERE combinada con el operador NOT: WHERE NOT department = 'Information Technology' ;. El operador NOT invierte la condición, filtrando y devolviendo rápidamente los perfiles de todos los empleados regulares de la organización que no pertenecen a Tecnologías de la Información.

```
MariaDB [organization]> SELECT *  
-> FROM employees  
-> WHERE NOT department = 'Information Technology';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434
1003	d394e816f943	sgilmore	Finance	South-153
1004	e218f877g788	eraab	Human Resources	South-127
1005	f551g340h864	gesparza	Human Resources	South-366
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1015	p611q262r945	jsoto	Finance	North-271
1016	q793r736s288	sbaelish	Human Resources	North-229
1017	r550s824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1020	u899v381w363	arutley	Marketing	South-351
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	iuduik	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1026	a998b568c863	apatel	Human Resources	West-320
1027	b806c503d354	mrah	Marketing	West-246
1028	c603d749e374	aestrada	Human Resources	West-121
1029	d336e475f676	ivelasco	Finance	East-156
1030	e391f189g913	mabadi	Marketing	West-375
1031	f419g188h578	dkot	Marketing	West-408
1034	i679j565k940	bsand	Human Resources	East-484
1035	j236k303l245	bisles	Sales	South-171
1036	k550l533m205	rjensen	Marketing	Central-239
1038	m873n636o225	btang	Human Resources	Central-260

- **Figura 6.** Filtrado general del personal de la organización excluyendo al departamento de Tecnologías de la Información.

Resumen

En esta actividad, apliqué filtros a consultas SQL para obtener información específica sobre intentos de inicio de sesión y equipos de empleados. Trabajé con dos tablas diferentes: `log_in_attempts` y `employees`. Utilicé los operadores lógicos AND, OR y NOT para detallar las condiciones exactas de cada tarea de seguridad. Además, empleé el operador LIKE junto con el comodín de porcentaje (%) para filtrar patrones de texto. Estas acciones permiten extraer inteligencia precisa de los registros para auditar accesos y gestionar permisos de manera integral.

Apply filters to SQL queries

Apply filters to SQL queries

Project description

This project demonstrates the use of SQL (Structured Query Language) queries to analyze system logs and investigate potential security incidents. The fundamental goal is to apply logical filters to audit user access and manage organizational information, ensuring that anomalies can be detected efficiently. By mastering these queries, we improve monitoring capabilities, facilitate incident response, and strengthen control over critical database information.

Retrieve failed login attempts outside of business hours

To begin the investigation of potential anomalous activities, I extracted records of failed access attempts occurring after business hours (18:00). First, I used `SELECT *` to select all data from the `log_in_attempts` table. Then, I applied a `WHERE` clause with an `AND` operator to filter the results. The first condition, `login_time > '18:00'`, filters attempts after 18:00, and the second condition, `success = FALSE`, isolates failed logins. This ensures precision when investigating high-risk events.

```
MariaDB [organization]> SELECT *  
-> FROM log_in_attempts  
-> WHERE login_time > '18:00' AND success = FALSE;
```

event_id	username	login_date	login_time	country	ip_address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
52	cjackson	2022-05-10	22:07:07	CAN	192.168.58.57	0
69	wjaffrey	2022-05-11	19:55:15	USA	192.168.100.17	0
82	abernard	2022-05-12	23:38:46	MEX	192.168.234.49	0
87	apatel	2022-05-08	22:38:31	CANADA	192.168.132.153	0
96	ivelasco	2022-05-09	22:36:36	CAN	192.168.84.194	0
104	asundara	2022-05-11	18:38:07	US	192.168.96.200	0
107	bisles	2022-05-12	20:25:57	USA	192.168.116.187	0
111	aestrada	2022-05-10	22:00:26	MEXICO	192.168.76.27	0
127	abellmas	2022-05-09	21:20:51	CANADA	192.168.70.122	0
131	bisles	2022-05-09	20:03:55	US	192.168.113.171	0
155	cgriffin	2022-05-12	22:18:42	USA	192.168.236.176	0
160	jclark	2022-05-10	20:49:00	CANADA	192.168.214.49	0
199	yappiah	2022-05-11	19:34:48	MEXICO	192.168.44.232	0

19 rows in set (0.015 sec)

Figure 1. Visualization of failed login attempts occurring after 18:00 hours using the `AND` clause.

Retrieve login attempts on specific dates

In order to audit system activity during a potential incident, I reviewed logins for two specific days. I started by selecting all data from the `log_in_attempts` table. Then, I used the WHERE clause with the OR operator to filter the query: `WHERE login_date = '2022-05-09' OR login_date = '2022-05-08'`; . The first condition filters access from 2022-05-09 and the second from 2022-05-08. The OR operator is indispensable here because it returns records that match at least one of the specified dates.

```
MariaDB [organization]> SELECT *  
-> FROM log_in_attempts  
-> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
24	arusso	2022-05-09	06:49:39	MEXICO	192.168.171.192	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1
26	apatel	2022-05-08	17:27:00	CANADA	192.168.123.105	1
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
30	yappiah	2022-05-09	03:22:22	MEX	192.168.124.48	1
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
36	asundara	2022-05-08	09:00:42	US	192.168.78.151	1
38	sbaelish	2022-05-09	14:40:01	USA	192.168.60.42	1
39	yappiah	2022-05-09	07:56:40	MEXICO	192.168.57.115	1
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
43	mcouliba	2022-05-08	02:35:34	CANADA	192.168.16.208	0
44	daquino	2022-05-08	07:02:35	CANADA	192.168.168.144	0
47	dkot	2022-05-08	05:06:45	US	192.168.233.24	1
49	asundara	2022-05-08	14:00:01	US	192.168.173.213	0
53	nmason	2022-05-08	11:51:38	CAN	192.168.133.188	1
56	acook	2022-05-08	04:56:30	CAN	192.168.209.130	1
58	ivelasco	2022-05-09	17:20:54	CAN	192.168.57.162	0
61	dtanaka	2022-05-09	09:45:18	USA	192.168.98.221	1
65	aalonso	2022-05-09	23:42:12	MEX	192.168.52.37	1
66	aestrada	2022-05-08	21:58:32	MEX	192.168.67.223	1
67	abernard	2022-05-09	11:53:41	MEX	192.168.118.29	1
68	mrh	2022-05-08	17:16:13	US	192.168.42.248	1
70	tmitchel	2022-05-09	10:55:17	MEXICO	192.168.87.199	1
71	mcouliba	2022-05-09	06:57:42	CAN	192.168.55.169	0
72	alevitsk	2022-05-08	12:09:10	CANADA	192.168.139.176	1

Figure 2. Extraction of access logs for consecutive dates using the OR logical operator.

Retrieve login attempts outside of Mexico

As part of the international access monitoring, I performed filtering to identify traffic originating outside of Mexico. I started by selecting all data from `log_in_attempts`. Afterwards, I used a WHERE clause with the NOT operator to filter countries that are not Mexico: `WHERE NOT country LIKE 'MEX%'`; I used LIKE with the pattern MEX% because the database represents Mexico as MEX and MEXICO. The percentage sign (%) works as a wildcard to cover any number of characters, thus effectively excluding all traffic from that region.

```
MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE NOT country LIKE 'MEX%';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
7	eraab	2022-05-11	01:45:14	CAN	192.168.170.243	1
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
10	jrafael	2022-05-12	09:33:19	CANADA	192.168.228.221	0
11	sgilmore	2022-05-11	10:16:29	CANADA	192.168.140.81	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
13	mrhah	2022-05-11	09:29:34	USA	192.168.246.135	1
14	sbaelish	2022-05-10	10:20:18	US	192.168.16.99	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
17	pwashing	2022-05-11	02:33:02	USA	192.168.81.89	1
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
19	jhill	2022-05-12	13:09:04	US	192.168.142.245	1
21	iuduik	2022-05-11	17:50:00	US	192.168.131.147	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1
26	apatel	2022-05-08	17:27:00	CANADA	192.168.123.105	1
29	bisles	2022-05-11	01:21:22	US	192.168.85.186	0
31	acook	2022-05-12	17:36:45	CANADA	192.168.58.232	0
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
33	zbernal	2022-05-11	02:52:10	US	192.168.72.59	1
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
36	asundara	2022-05-08	09:00:42	US	192.168.78.151	1
37	eraab	2022-05-10	06:03:41	CANADA	192.168.152.148	0
38	sbaelish	2022-05-09	14:40:01	USA	192.168.60.42	1
41	apatel	2022-05-10	17:39:42	CANADA	192.168.46.207	0
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0

Figure 3. Filtering of login events excluding traffic originated in Mexico using the NOT and LIKE operators.

Retrieve employees in Marketing

To update the equipment for Marketing staff in the East region, I needed to isolate these employees. I selected the data from the employees table and used the WHERE clause with the AND operator: WHERE department = 'Marketing' AND office LIKE 'East%';. The first condition, department = 'Marketing', filters for staff in that department. The second condition, office LIKE 'East%', uses the % wildcard to match any office starting with the word "East" (e.g., East-170 or East-195), providing an exact listing.

```
MariaDB [organization]> SELECT *  
-> FROM employees  
-> WHERE department = 'Marketing' AND office LIKE 'East%';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1052	a192b174c940	jdarosa	Marketing	East-195
1075	x573y883z772	fbautist	Marketing	East-267
1088	k865l965m233	rgosh	Marketing	East-157
1103	NULL	randerss	Marketing	East-460
1156	a184b775c707	dellery	Marketing	East-417
1163	h679i515j339	cwilliam	Marketing	East-216

7 rows in set (0.000 sec)

Figure 4. Visualization of Marketing department staff located in the East region offices.

Retrieve employees in Finance or Sales

To deploy a specific security update, I required the list of users from the Finance and Sales areas. I began by querying the employees table. Then, I applied the WHERE clause with the OR operator: `WHERE department = 'Finance' OR department = 'Sales'`; I used OR instead of AND because I need employees belonging to either of the two departments. The first condition filters those in Finance (`department = 'Finance'`) and the second those in Sales (`department = 'Sales'`), effectively unifying the list.

```
MariaDB [organization]> SELECT *  
-> FROM employees  
-> WHERE department = 'Finance' OR department = 'Sales';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1015	p611q262r945	jsoto	Finance	North-271
1017	r550s824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	iuduike	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1029	d336e475f676	ivelasco	Finance	East-156
1035	j236k303l245	bisles	Sales	South-171
1039	n253o917p623	cjackson	Sales	East-378
1041	p929q222r778	cgriffin	Sales	North-208
1044	s429t157u159	tbarnes	Finance	West-415
1045	t567u844v434	pwashing	Finance	East-115
1046	u429v921w138	daquino	Finance	West-280
1047	v109w587x644	cward	Finance	West-373
1048	w167x592y375	tmitchel	Finance	South-288
1049	NULL	jreckley	Finance	Central-295
1050	y132z930a114	csimmons	Finance	North-468
1057	f370g535h632	mscott	Sales	South-270
1062	k367l639m697	redwards	Finance	North-180
1063	l686m140n569	lpope	Sales	East-226
1066	o678p794q957	ttyrell	Sales	Central-444
1069	NULL	jpark	Finance	East-110
1071	t244u829v723	zdutchma	Sales	West-348
1072	u905v920w694	esmith	Sales	East-421

Figure 5. Combined extraction of employee profiles corresponding to the Finance and Sales departments.

Retrieve all employees who are not in IT

For a security update targeted at users outside of IT, I needed to exclude staff from that department. After selecting data from the employees table, I used the WHERE clause combined with the NOT operator: `WHERE NOT department = 'Information Technology'`; . The NOT operator inverts the condition, quickly filtering and returning the profiles of all regular employees in the organization who do not belong to Information Technology.

```
MariaDB [organization]> SELECT *  
-> FROM employees  
-> WHERE NOT department = 'Information Technology';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434
1003	d394e816f943	sgilmore	Finance	South-153
1004	e218f877g788	eraab	Human Resources	South-127
1005	f551g340h864	gesparza	Human Resources	South-366
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1015	p611q262r945	jsoto	Finance	North-271
1016	q793r736s288	sbaelish	Human Resources	North-229
1017	r550s824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1020	u899v381w363	arutley	Marketing	South-351
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	iuduike	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1026	a998b568c863	apatel	Human Resources	West-320
1027	b806c503d354	mrah	Marketing	West-246
1028	c603d749e374	astrada	Human Resources	West-121
1029	d336e475f676	ivelasco	Finance	East-156
1030	e391f189g913	mabadi	Marketing	West-375
1031	f419g188h578	dkot	Marketing	West-408
1034	i679j565k940	bsand	Human Resources	East-484
1035	j236k303l245	bisles	Sales	South-171
1036	k550l533m205	rjensen	Marketing	Central-239
1038	m873n636o225	btang	Human Resources	Central-260

- **Figure 6.** General filtering of organization staff excluding the Information Technology department.

Summary

In this activity, I applied filters to SQL queries to obtain specific information about login attempts and employee teams. I worked with two different tables: `log_in_attempts` and `employees`. I used the logical operators AND, OR, and NOT to detail the exact conditions for each security task. Additionally, I used the LIKE operator along with the percentage wildcard (%) to filter text patterns. These actions allow for the extraction of precise intelligence from logs to audit access and manage permissions comprehensively.